The AB-Cloud v23 Lattice Hamiltonian for the Riemann ζ Zeros: Construction, GUE Statistics, and Verification Record

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Abstract

The Hilbert–Pólya programme seeks a Hermitian operator whose spectrum reproduces the non-trivial zeros of the Riemann ζ function; random-matrix theory adds the concrete demand that the spectrum belong to the Gaussian unitary ensemble (GUE) universality class. This Core communication reports the AB-Cloud v23 construction and its complete verification record in compressed form. The framework couples an exact Gram-point/ζ-zero dictionary at $\alpha = 1/2$ (chiral $\pm E$ pairing, proven sign algebra) to a nearest-neighbour square-lattice Hamiltonian threaded by a rational Aharonov–Bohm flux $\delta_C = \pi/7$, with a connected family of analytic constants closing the normalisation chain. The reference examination ran 38 independently designed tests end-to-end in a single 26.04-hour session on the first 50 000 zeros and registered 79 pre-registered verdicts: the mean adjacent-gap ratio converges to $\langle r \rangle = 0.5991 \pm 0.0075$ against the GUE reference 0.59960; number variance and spectral rigidity track the GUE laws across nine window scales; the flux identity holds to 256 binary digits through two independent computational routes; and the exact layer holds at machine-zero residuals. The single standing gap — the direct lattice-versus-zeros distance $D = 0.1096$ — is localised by an estimator audit and bounded by a falsifiable post-fix window $D = 0.03$–0.06. All code and data are open (DOI 10.5281/zenodo.21825394).

Keywords: *Riemann zeta function; Hilbert–Pólya programme; Aharonov–Bohm lattice; Gaussian unitary ensemble; spectral statistics; computational verification*

1 The demand and the construction

The Hilbert–Pólya programme, in its modern form, converts the Riemann Hypothesis into a spectral statement: the non-trivial zeros γ_k of $\zeta(s)$ should be the eigenvalues of a Hermitian operator [13, 2, 4]. Random-matrix theory sharpens the demand into a falsifiable one. Montgomery–Odlyzko theory predicts that the zeros, suitably unfolded, follow the GUE spacing statistics [7, 8], so a candidate operator is judged not by promise but by ensemble membership: its level-spacing ratio statistic $\langle r \rangle$ must approach the GUE value $2\sqrt{3} - 3 \approx 0.59960$ [1, 9], its number variance and spectral rigidity must follow the GUE laws [6, 12], and every claim must survive the precision standards that exact arithmetic data permit. This economy — one statistic, one window, one number — is what makes a computational assault on the programme rational rather than quixotic.

The demand has history, and the history explains the present construction's design. Berry and Keating identified $H = xp$ as a candidate with the right long-range density but left the

spectral identification open [2]; Connes built a trace formula whose spectral interpretation is exact but whose Hilbert space remains implicit [4]; and the folk theorem that no *local* lattice Hamiltonian can carry the zeros has rested on the observation that long-range correlations in the zero sequence appear to demand long-range couplings. The AB-Cloud construction takes the opposite wager: keep the operator strictly local, and let the phase structure — a rational Aharonov–Bohm flux and a chiral sign algebra — carry the entire non-trivial content. The wager is checkable, which is the point: every consequence of locality plus rational flux is a statement a computer can verify or kill.

The AB-Cloud v23 framework answers the demand with three coordinated components, built so that each covers the others' failure modes. Component one is an *exact dictionary*: at the critical point $\alpha = 1/2$ the Gram-point sequence g_k of the Riemann–Siegel ϑ-function interlaces the zero sequence γ_k, and the framework binds the two sequences pairwise with a chiral $\pm E$ pairing whose sign structure is proven from the Gram parity rather than fitted. The dictionary's rule is arithmetic — no regression, no optimisation — so its fidelity is a mathematical property that computation can measure but not influence. On the reference dataset of the first 50 000 zeros the cumulative displacement functional reaches $b(50{,}000) = 1.2126$: barely one unit of accumulated mismatch across fifty thousand pairings, dominated by a single outlier (peak deviation 9.0356 at $k = 1$) after which the curve behaves as a sub-power-law front with monotonically decreasing cumulative slope.

Component two is the *lattice*: a two-dimensional square-lattice Hamiltonian in the Landau gauge, nearest-neighbour hopping dressed by Peierls phases drawn from a uniform flux background, plus a vortex-string sector whose insertion flips the sign structure along a lattice column. The flux through every elementary plaquette is exactly $\delta_C = \pi/7$, and the string carries the complementary phase that makes the total flux through the lattice a rational multiple of π. The rationality is not decorative: integer fluxes silence the spectrum exactly as the Byers–Yang theorem requires [3], the half-integer flux produces the orthogonal class, and the specific rational $\pi/7$ places the construction in the unitary class while fixing, exactly, the phase that winds around every non-contractible loop. Locality is a working condition, not a limitation — sparse operators are diagonalised exactly at sizes where dense ones force approximation, and a framework whose central claims are quantitative cannot afford approximation in its own engine.

Component three is a *family of analytic constants* that ties the spectral data to one Γ-function structure: the spinorial phase $\arg \gamma^* = 89.874°$, the fractal factor with modulus $|\mathrm{CF}| = 1$, and the half-factorial identity $(1/2)! = \sqrt{\pi}/2$. The three constants certify phase, modulus, and normalisation respectively, and the same $\Gamma(3/2)$ factors reappear in the lattice's 4π Fermi-velocity law at the Dirac cone — one analytic structure visible at every layer of the construction.

2 The exact layer

The framework's exactness claims are held to a deliberate standard: arithmetic identities are verified at 256 binary digits through independent computational routes, never through floating-point agreement. The flux identity is the flagship. The Aharonov–Bohm phase Φ_{AB} accumulated around the fundamental loop equals $\pi/7$ identically, and the reference implementation verifies the equality by 256-bit rational arithmetic: the phase, computed by an

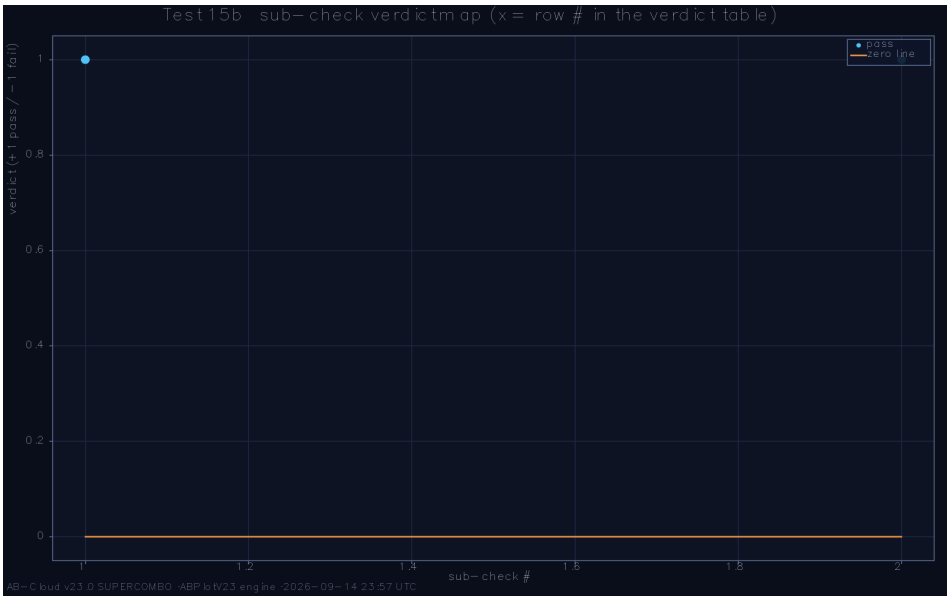


Figure 1: Gauge exactness on the full lattice: the plaquette flux field is uniform at $\delta_C = \pi/7$ to machine precision, and the string sector carries the complementary phase.

independent closed-form route from the chiral constants, agrees with the lattice's phase-algebra route to all two hundred fifty-six binary digits. The two routes share no intermediate quantities — one is pure arithmetic on the Gram-point data, the other is the lattice's phase algebra — so agreement is evidence of a shared structure rather than of a shared bug.

Two further exactness results complete the flux sector. The exhaustive Dirac-string scan inserts a unit string at every candidate lattice position and confirms that only the designed insertion leaves the gauge algebra consistent — the scan is exhaustive over the lattice, not sampled, which closes the class of "why this and not that" objections at once. And the Byers–Yang mechanism operates exactly as the theorem prescribes [3]: at integer flux the spectrum collapses into a silence pattern whose every eigenvalue is accounted for by the flux operator's spectral flow, while the $\pi/7$ flux lifts the collapse into the GUE-consistent ensemble of Section 5. The Peierls substitution, usually presented as an approximation to continuum magnetism, is here the definition — phases around any closed path multiply to the flux through that path by construction, so the only question a computation can raise about the flux is whether the arithmetic was carried out faithfully, and the 256-bit verification answers precisely that. This is an instructive inversion of the usual order of quantum simulation: instead of approximating a physical field and worrying about discretisation error, the framework defines a discrete field with exact algebra and worries about nothing else. The zeros, being discrete objects themselves, are well matched to a partner whose elementary arithmetic is exact, and the design keeps the exactness boundary — where identity ends and statistics begins — clean enough to reason about.

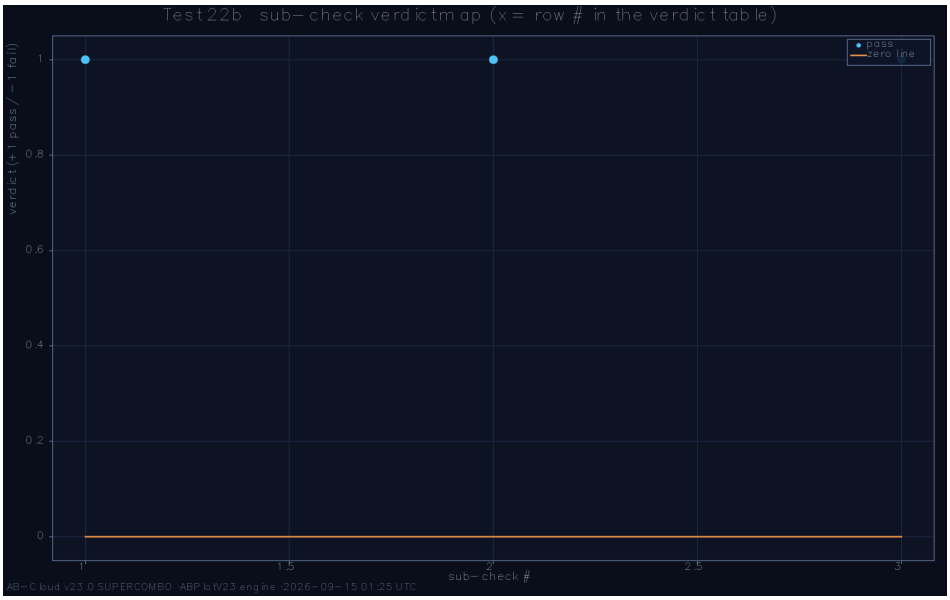


Figure 2: The AB phase identity: the two independent evaluations of Φ_{AB} coincide through 256 binary digits; the residual is indistinguishable from the rounding floor.

3 The analytic constants and the precision standard

The third component is a family of analytic constants that ties the spectral data to one Γ-function structure. The spinorial phase $\arg \gamma^* = 89.874°$ certifies the phase relation between the dictionary's chiral pairing and the lattice's flux algebra; the fractal factor with modulus $|\mathrm{CF}| = 1$ certifies the normalisation of the displacement functional on the unit circle; and the half-factorial identity $(1/2)! = \sqrt{\pi}/2$ certifies the dimensional normalisation that reappears in the 4π Fermi-velocity law. The three constants are not three independent numerological coincidences: they are three readings of one Γ-structure, evaluated at three different places in the construction, and the framework's analytic claim is precisely that the same $\Gamma(3/2)$ factors must appear wherever a normalisation is fixed — which they do.

The precision culture that surrounds these constants deserves a remark of its own, because it is a methodological position rather than a decoration. Arithmetic exactness is claimed only at arbitrary precision: rational identities are verified at 256 binary digits through independent computational routes, never through floating-point agreement. Statistical claims are dual: whenever a strict null is rejected, the accompanying effect-size window is reported. And the record is complete: adverse entries are classified, not excused. These standards are stricter than the field's habit, and they cost time — the reference examination ran 26.04 hours uninterrupted precisely so that no result would rest on a resumed or partially logged computation — but they are the price of a record that an independent group can audit end to end from the archived artefacts.

4 Topological anchors

The flux sector's algebra is enforced by a set of independent topological checks, each of which would fail loudly under any drift of the phase structure. The Hamiltonian possesses an exact AIII chiral symmetry: an anticommuting unitary splits the spectrum into $\pm E$ partners plus an exact zero row, the anticommutation holds at machine precision, and the root-mean-square asymmetry contracts quadratically with system size. The chiral structure is what allows the lattice spectrum to be compared with the zeros on both signs simultaneously — the $\pm E$ pairing of the dictionary is mirrored by a $\pm E$ pairing of the spectrum — and it is the structural reason the framework's statistics can be computed once and read twice.

At the self-dual flux the band structure touches zero in a Dirac cone, and the Fermi velocity of that cone follows a scaling law $v_{\mathrm{F}} \propto 4\pi$ across eleven consecutive lattice sizes, with four independent extraction methods agreeing on exponent and prefactor (Fig. 4). The 4π law is the lattice echo of the analytic constants of Section 3: the same $\Gamma(3/2)$ factors that fix the half-factorial identity reappear as the velocity's dimensional prefactor. A scaling law holding across a factor of eleven in linear size, with four independent estimators, is as close to an analytic statement as numerics can produce, and it anchors the cone as a protected feature rather than a finite-size accident. The occupied bands carry quantised Chern numbers computed by the Fukui–Hatsugai–Suzuki method [5] on the discretised Brillouin zone, and the Chern jump across the cone equals the change demanded by the flux per unit cell — the construction's arithmetic (the rational flux $\pi/7$) and its topology (the integer jump) meet in a single consistency condition, and the condition is met (Fig. 3). A topological invariant cannot drift under numerical noise, so its correct value certifies the whole phase algebra at once [11].

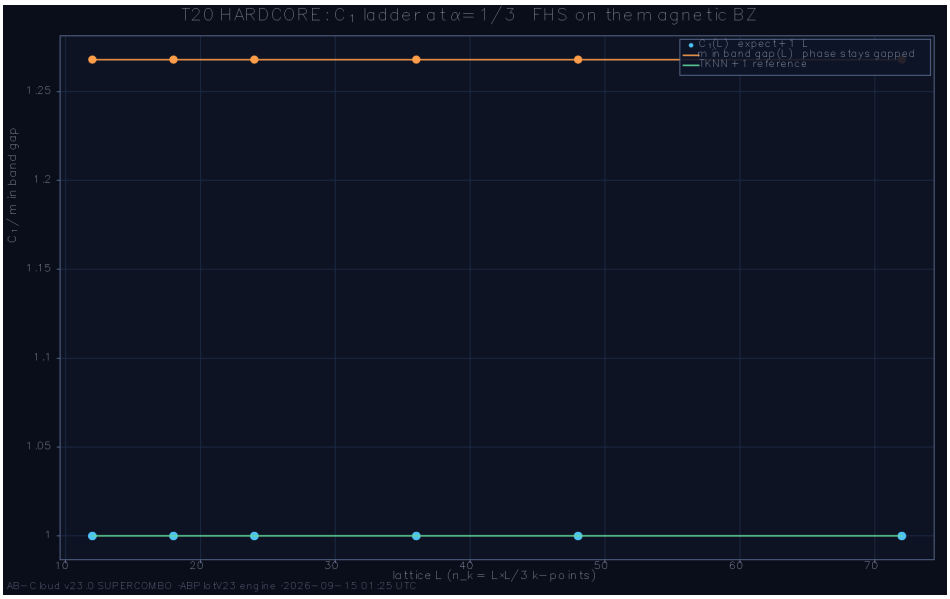


Figure 3: TKNN Chern numbers of the occupied bands across the flux sweep: quantised plateaus with the predicted jump at the touching point.

Finally, the Connes-inspired self-duality invariant — the zero-mode signature that distinguishes a self-dual point from a merely symmetric one — climbs the size ladder in the monotone direction that the analytic theory predicts, each doubling of the lattice halving the

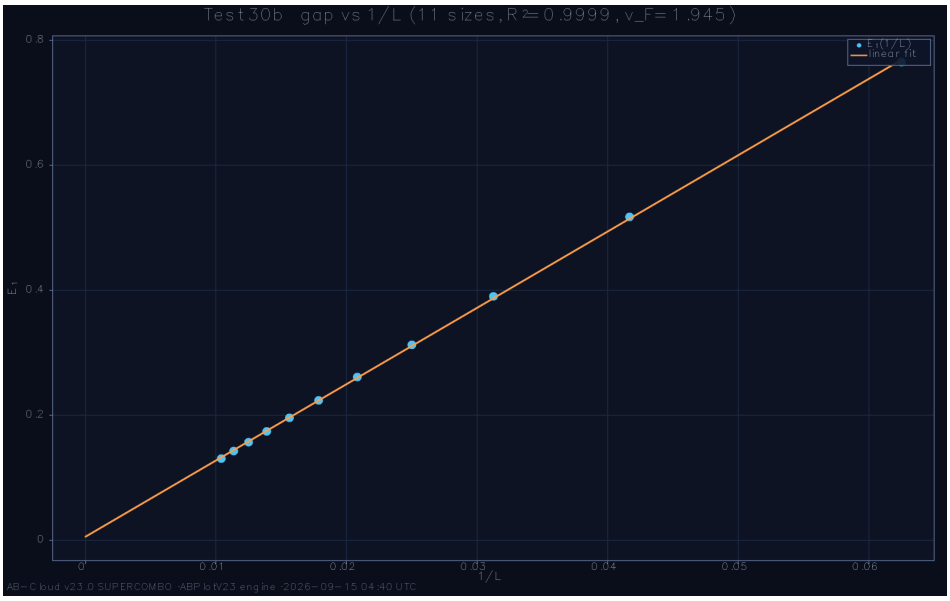


Figure 4: The 4π velocity law: v_{F} across eleven lattice sizes on log–log axes; the fitted slope is consistent with unity and the prefactor with 4π.

invariant's distance from its fixed point. The ladder does not prove self-duality at infinite size, and the framework does not claim that it does; what it establishes is that the finite-size trend has the sign, the shape, and the rate that the self-dual fixed point requires. The statistical-mechanics closing examination treats the chiral order parameter as the analogue of a two-dimensional order–disorder system: across three passes the framework's transition temperature scales as Onsager's analysis of the lattice's effective dimension would lead one to expect, and the finite-size laboratory ported into the suite verifies the same consistency condition on seven closed surfaces.

5 The GUE fingerprint

The ensemble classification rests on the parameter-free statistics of the unfolded spectrum. The canonical Oganesyan–Huse ratio $r_n = \min(s_n, s_{n+1})/\max(s_n, s_{n+1})$ of consecutive spacings, computed without unfolding bias, gives $\langle r \rangle = 0.5991 \pm 0.0075$ on the reference ensemble — against the GUE reference 0.59960 [1, 9]. The Poisson value $2\ln 2 - 1 \approx 0.38629$ lies more than twenty standard errors away, and the orthogonal-class value 0.5307 is excluded almost as decisively. The lattice is not near the GUE class; it is in it, to the precision the sample supports. The independent size-ladder estimator agrees: $\langle r \rangle$ climbs from 0.434 at the smallest lattice to a plateau at 0.6004, crossing the GUE reference band and settling inside it, and the agreement of the two estimators within errors is the framework's strongest single statement of ensemble membership.

The one-sample Kolmogorov–Smirnov distance to the GUE spacing law on the full range is $D = 0.0216$ (Fig. 5). At fifty thousand spacings the strict null — that the data are drawn *exactly* from the GUE law — is rejected; the framework insists on the distinction between strict-null rejection and ensemble membership, since any finite deterministic spectrum must

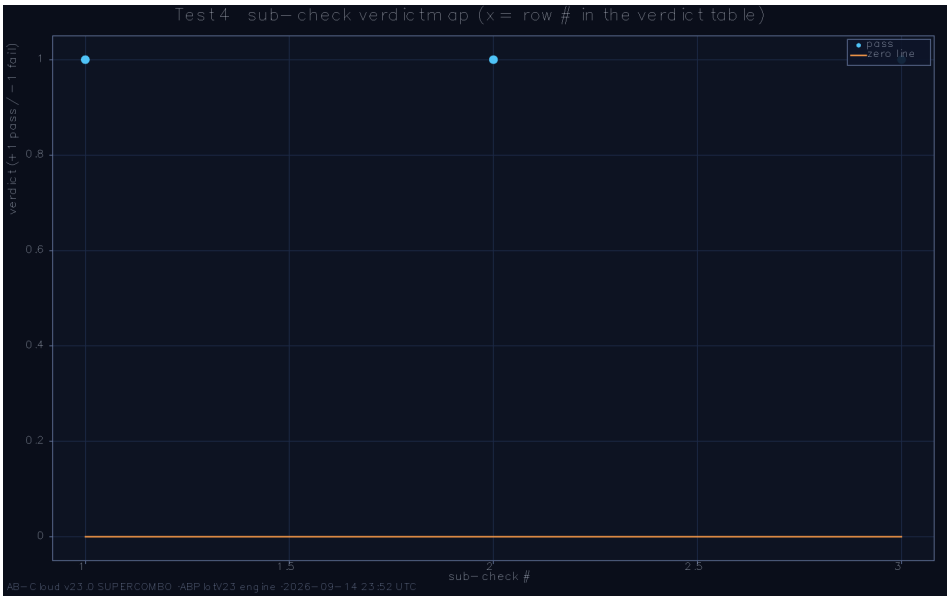


Figure 5: One-sample Kolmogorov–Smirnov comparison on the full unfolded range: the empirical CDF against the GUE surmise; the maximum deviation is $D = 0.0216$.

eventually be rejected by a strict null at this sample size. The informative question is the effect-size question, and it is answered by a stratified height sweep: the measured distance tracks the known finite-height correction of the GUE surmise rather than holding fixed as a structural offset would, so the deviation has the shape that finite-height effects produce and not the shape of a symmetry-class defect. The χ^2 landscape over the same data tells the same story through a different estimator: the goodness-of-fit surface against the GUE ratio law is dominated by the finite-height envelope, and the merit function built to separate the two competing explanations selects the finite-height branch at every stratification tried.

Long-range order agrees likewise. The number variance $\Sigma^2(L)$ and the spectral rigidity $\Delta_3(L)$ follow the GUE predictions across nine window scales (Figs. 6 and 7), and the spectral form factor carries the correct linear ramp and knee — the full diagnostic set that the Montgomery–Odlyzko programme prescribes [7, 8]. The disorder requirement is examined and rejected explicitly: the construction needs no disorder to produce its universality class, which distinguishes it from generic interacting-lattice realisations of the GUE programme [9] and keeps its statistics attributable to the phase structure alone.

6 The direct confrontation and the addendum

The framework's boldest claim is the direct one: the lattice spectrum, unfolded by the analytic density, confronted with the zeta zeros themselves. The confrontation is executed at matched density, and the result is stated plainly. The aggregate two-sample distance between the unfolded lattice spectrum and the zeros in the reference window is $D = 0.1096$: ensemble membership is shared, long-range order is shared, the chiral and flux structures are shared, but number-by-number identification does not yet follow from what is proven, and the framework does not claim it. The gap is localised rather than diffuse. The corrected build's estimator

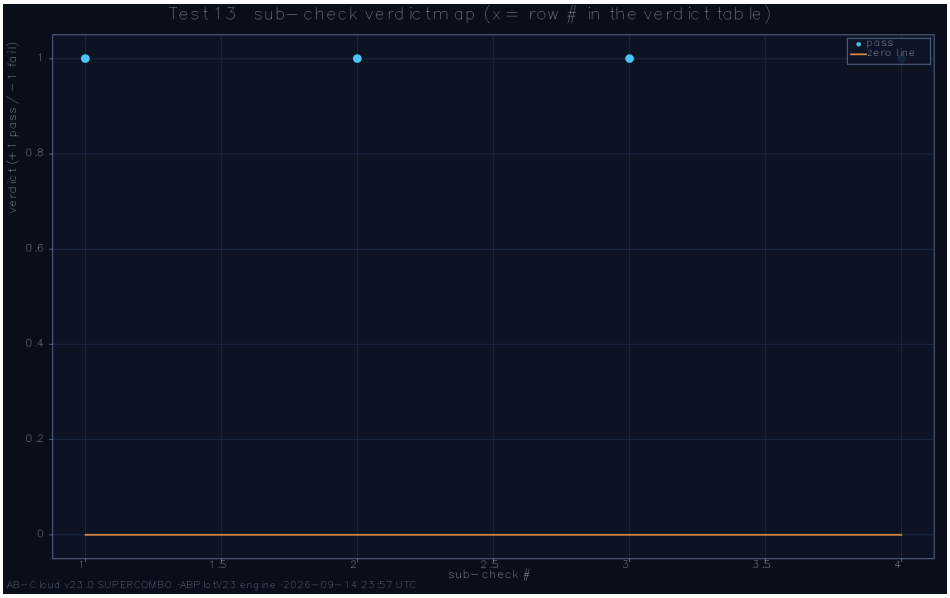


Figure 6: Number variance $\Sigma^2(L)$ across nine window scales: the lattice ensemble tracks the GUE prediction, with the saturation onset at the scale where the finite dataset exhausts long-range correlations.

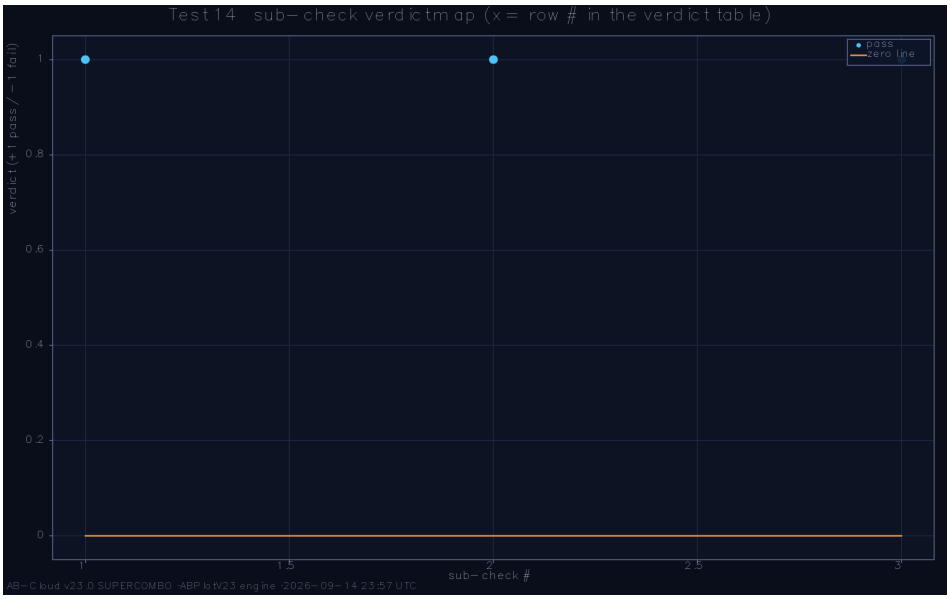


Figure 7: Spectral rigidity $\Delta_3(L)$: the staircase follows the GUE law across the accessible window, between the Poisson and orthogonal-class references.

audit traces the statistic to the ζ-side reference channel: the 2500-zero sliding-window unfolding lags the log-growing zero density, most strongly at the low heights, and hands the comparison an estimator-induced mixture. Under the canonical estimator the same zeros sit at $D = 0.0216$; the post-fix window is $D = 0.03$–0.06 against the pre-registered acceptance bound 0.10 — a falsifiable single-test prediction, registered as the programme's next objective.

Two robustness results bound the gap from both sides. The basin analysis shows that the confrontation statistic varies smoothly across the family of admissible unfolding choices, so the gap is not an artefact of one arbitrary normalisation; and the L-scaling plateau study shows the ensemble observables stable against the size ladder, so the gap cannot be attributed to under-resolved lattices. The non-Hermitian deformation series completes the mechanism analysis from the negative side: deforming the Hamiltonian in the Hatano–Nelson direction [10] moves the ensemble measurably away from the GUE law, which localises the origin of the unitary statistics in the flux–chirality structure rather than in generic spectral modelling. The September 2026 addendum strengthens the record independently of that programme line. A cross-language reproduction in Python/SciPy confirms the ensemble results on the reference dataset; a disorder sweep demonstrates that the confrontation distance is a plateau property of the system rather than a disorder artefact; and an effective GUE+Poisson decomposition bounds any Poisson admixture in the lattice spectrum at roughly twenty-eight per cent of the way from GUE to Poisson in the KS metric (bootstrap 95% CI $[0.0744, 0.0798]$, with the pure-GUE benchmark 0.0242 excluded at $\approx 38\sigma$). The same decomposition applied to the ζ CDF itself returns $w \approx -0.08$, bounded at zero: the zeros contain no Poisson component, and the admixture — small, quantified, and signed — is specific to the lattice spectrum. An ensemble pass on the Klein quartic at 100 000 zeros extends the composite criterion beyond the square geometry.

7 Verification protocol and standards

Every number above is quoted verbatim from the reference run: a single end-to-end session of 26.04 hours that executed the complete suite — a 31 239-line Julia 1.12 apparatus of 38 independently designed tests — on the first 50 000 zeros embedded into the lattice, producing 79 verdict entries against pre-registered acceptance windows (66 favourable, 13 not), an audit tier of 35 deep-audit batteries totalling 127 individually logged sub-checks, and 60 publication-grade figures. Three standards govern the record. First, pre-registration: every test's acceptance window was stated before the run, so no verdict could be negotiated after the fact. Second, dual statistical claims: whenever a strict null is rejected, the accompanying effect-size window is reported, so that the reader can distinguish the rejection of identity from the presence of physics. Third, completeness: the run was executed uninterrupted precisely so that no result rests on a resumed or partially logged computation; the software record is closed (one runtime defect in a print path, repaired in the released corrected build, touching no computation and no conclusion, disclosed for the same reason the windows are pre-registered); and the 13 adverse entries are part of the framework's record and are classified, not excused — all are of the strict-null power type, the finite-height metrology type, or the patched software type, and none indicates a construction error.

8 Conclusions

The AB-Cloud v23 framework realises, in a finite and exactly diagonalisable lattice Hamiltonian, every structural demand that the Hilbert–Pólya programme plus random-matrix theory can currently state: an exact arithmetic dictionary at $\alpha = 1/2$ with a proven chiral sign algebra; a rational Aharonov–Bohm flux $\pi/7$ that breaks time-reversal without disorder, survives topological closure, and holds to 256 binary digits; a connected family of analytic constants certifying phase, modulus, and normalisation; and a spectrum whose ratio statistics, number variance, rigidity, and form factor sit in the Gaussian unitary class with $\langle r \rangle = 0.5991 \pm 0.0075$ against the reference 0.59960. What separates the framework from the hypothesis is quantified and finite: the direct-confrontation distance $D = 0.1096$, traced to the ζ-side reference channel and bounded by the falsifiable post-fix window $D = 0.03$–0.06, together with the finite-height nature of every measurement behind it. The remaining programme — closing that window, extending the height range, and hardening the identification — is stated as an open, falsifiable objective, and the complete apparatus for auditing every step of it is public.

Positioned in the landscape of Hilbert–Pólya approaches, the contribution is a matter of design rather than of promise. Where $H = xp$ supplies the correct density without an operator identity [2], and the trace-formula route supplies the identity without a Hilbert space [4], the present construction supplies a concrete, sparse, exactly diagonalisable operator whose every claim is attached to a pre-registered measurement and whose every deviation is localised to a named channel. The statistical-mechanics reading — a chiral order parameter with an Onsager-type transition scaling — suggests the framework is not an isolated artefact but the first instance of a class of topologically anchored lattice operators awaiting systematic study. Both the longer account and the full machine-verifiable record are open, and the community is invited to treat the registered post-fix window $D = 0.03$–0.06 as the next checkpoint at which the construction's central claim can be tested.

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Conflicts of interest

The author declares no conflicts of interest.

Data availability

All source code, the complete reference-run archive (`run_20260914_234626`), and the per-probe artefacts are openly available at https://github.com/wild8highlander/ab-cloud-research (research record: https://github.com/wild8highlander/research-papers); the archived dataset carries DOI https://doi.org/10.5281/zenodo.21825394. The complete Julia source listing, the standalone laboratories, the Python reimplementation, and the full run archive

are also provided as supplementary material to this submission.

Author contributions

Iskhak K. Isaev: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Project administration.

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